Class 11

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Background

We saw last dat that the main repository for biomolecular structure (the PDB database) only has ~250000 entries.

UniProtKB (the main protein sequence database) has over 200 million entries.

Alphafold

In this hands-on session we will utilize Alphafold to predict protein structure from sequence (Jumper et al. 2021).

Without the aid of such approaches, it can take years of expensive laboratory work to determine the structure of just one protein. With AlphaFold we can now accurately compute a typical protein structure in as little as ten minutes.

The EBI Alphafold Database

The EBI Alphafold database contains lots of computed structure models. It is increasingly likely that the structure you are interested in is already in this database at < https://alphafold.ebi.ac.uk/ >.

There are 3 major outputs from Alphafold.

1. A model of structure in **PDB** format.
2. a **plDDT** score: that tells us how confident the model is for a given residue in your protein (high values are good, above 70).
3. **PAE** score that tells us about protein packing ability.

If you can't find a matching sequence for the sequence you are interested in AFDB you can run Alphafold yourself…

Running Alphafold

We will use ColabFold to run Alphafold on our sequence at < https://github.com/sokrypton/ColabFold >.

Interperting Results

8. Custom analysis of resulting models In this section we will read the results of the more complicated HIV-Pr dimer AlphaFold2 models into R with the help of the Bio3D package. You can do the same thing for the monomer models if you wish but again they will be less interesting as the monomer is not physiologically relevant.

For tidiness we can move our AlphaFold results directory into our RStudio project directory. In this example my results are in the director results_dir (set below). You should change this to match your directory/folder name.

We can read all the Alphafold results into R and do more quantitative analysis than just viewing the structures in Mol-star:

Read all the PDB models:

```
library(bio3d)
pdbfiles <- list.files("hivpr_23119/", pattern=".pdb", full.names=T)
hivpbs <- pdbaln(pdbfiles, fit=T, exefile="msa")
```

```
Reading PDB files:
hivpr_23119/hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
hivpr_23119/hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
hivpr_23119/hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb
hivpr_23119/hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb
hivpr_23119/hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb
.....

Extracting sequences

pdb/seq: 1   name: hivpr_23119/hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4
pdb/seq: 2   name: hivpr_23119/hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1
pdb/seq: 3   name: hivpr_23119/hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5
pdb/seq: 4   name: hivpr_23119/hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2
pdb/seq: 5   name: hivpr_23119/hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3
```

```
#library(bio3dview)
#view.pdbs(hivpbs)
```

```
rd <- rmsd(hivpbs)
```

```
Warning in rmsd(hivpbs): No indices provided, using the 198 non NA positions
```

```
library(pheatmap)
colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```